

GCP VM Setup SOP — Elasticsearch + Kibana (Flipkart Internal)

Space: Retail-IP (RETAILIP) · Author: Anjali Jaiswal · Last modified: Jun 30, 2026

Source:

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Template Variables — Fill These In Before Following This SOP

This is a reusable template. Replace every `<placeholder>` with values from your own GCP project before running any command. Do not copy values from another team's setup.

Variable	Description	How to find
<code><your-gcp-project></code>	Your GCP project ID	GCP Console → Project selector
<code><your-vm-name></code>	Name for the VM you are creating	Choose a name relevant to your service
<code><your-zone></code>	GCP zone for the VM	Check your project's assigned region
<code><your-ssh-username></code>	Your Flipkart email with dots replaced by underscores	e.g. <code>firstname_lastname_flipkart_com</code>
<code><your-vm-internal-ip></code>	Internal IP assigned after VM creation	Output of the <code>gcloud instances create</code> command in Step 4
<code><an-existing-vm-in-your-project></code>	Any VM already running in your project	GCP Console → Compute Engine → VM instances
<code><your-cluster-name></code>	Name for your Elasticsearch cluster	Choose a name relevant to your project

Overview

This SOP documents the end-to-end steps to provision a GCP VM and install Elasticsearch 8.13.4 + Kibana 8.13.4. External internet is blocked on GCP VMs in the Flipkart network — all binaries must be transferred via SCP from a Mac that has internet access.

Key constraints in Flipkart GCP environment:

- No external internet on VMs

- Default VPC deleted — must use corporate VPC (look this up per-project, see Step 2)
 - Standard Debian images blocked by org policy — must use `fk-virt-infra` golden images
 - Old on-prem repo server `10.47.4.220` is unreachable from GCP subnets — do not reuse old Confluence setup scripts referencing on-prem repo IPs
 - Internal apt mirror `10.19.1.12` does not carry Elasticsearch packages
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Phase 1 — GCP VM Creation

Step 1: Set the project

```
gcloud config set project <your-gcp-project>
```

Step 2: Find the network config (reuse from an existing VM in your project)

```
gcloud compute instances describe <an-existing-vm-in-your-project> \
  --zone=<your-zone> \
  --project=<your-gcp-project> \
  --format="get(networkInterfaces[0].network, networkInterfaces[0].subnetwork)"
```

Flipkart deleted the default VPC — all VMs must use the corporate VPC. This command returns your project's actual VPC and subnet. Use whatever it outputs — do not hardcode another team's VPC or subnet name.

Step 3: Find the approved golden image

```
gcloud compute disks describe <an-existing-vm-in-your-project> \
  --zone=<your-zone> \
  --project=<your-gcp-project> \
  --format="get(sourceImage)"
```

Flipkart org policy `constraints/compute.trustedImageProjects` blocks `debian-cloud` images. Only images from the `fk-virt-infra` project are allowed. Use whichever golden image version this command returns — image versions are updated periodically.

Step 4: Create the VM

```
gcloud compute instances create <your-vm-name> \
  --project=<your-gcp-project> \
  --zone=<your-zone> \
  --machine-type=n2-standard-4 \
  --image=<golden-image-from-step-3> \
  --image-project=fk-virt-infra \
  --boot-disk-size=100GB \
  --boot-disk-type=pd-ssd \
  --network=<network-full-path-from-step-2> \
  --subnet=<subnet-full-path-from-step-2> \
```

```
--no-address \
--shielded-secure-boot \
--shielded-vtpm \
--shielded-integrity-monitoring \
--service-account=<your-project-default-compute-service-account> \
--
scopes=https://www.googleapis.com/auth/devstorage.read_only,https://www.googleapis.com/auth/logging.write,https://www.googleapis.com/auth/monitoring.write \
--tags=elasticsearch-node,vm-<your-region>
```

Note the internal IP assigned after creation — this becomes `<your-vm-internal-ip>` for all later steps.

Phase 2 — SSH Access Setup

Step 5: Add SSH key via OS Login

```
gcloud compute os-login ssh-keys add \
  --key-file=$HOME/.ssh/id_rsa.pub \
  --project=<your-gcp-project>
```

OS Login is enabled — GCP manages SSH identity via your Flipkart Google account. SSH username becomes your Flipkart email with dots replaced by underscores (e.g. `firstname_lastname_flipkart_com`). Use `$HOME` not `~` — the tilde can cause a "file not found" error.

Step 6: SSH into VM via jumphost

```
ssh -J global.jumphost-prod.fkcloud.in \
  -i ~/.ssh/google_compute_engine \
  <your-ssh-username>@<your-vm-internal-ip>
```

No external IP on the VM — direct SSH is blocked. Must route through Flipkart's jumphost.

If you see **"WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED"** — the jumphost key was rotated. Fix:

```
ssh-keygen -f "~/.ssh/known_hosts" -R "global.jumphost-prod.fkcloud.in"
```

Phase 3 — Elasticsearch Installation

Why not use internal repos? External internet is blocked on the VM. The old Flipkart repo server `10.47.4.220` is unreachable from GCP subnets (it lives on the old on-prem network). The internal apt mirror `10.19.1.12` has no ES packages. Old Confluence setup scripts referencing `10.47.4.220` do not work on GCP VMs. Solution: download on Mac, SCP to VM.

ES 5.x is incompatible with Java 17 — do not attempt. ES 5.x requires Java 8. Debian 12 ships Java 17. The `UseConcMarkSweepGC` JVM flag and `System::setSecurityManager` were both removed in Java 14+. Use ES 8.x directly.

Step 7: Download ES on Mac and SCP to VM

```
# Run on Mac (has internet access)
curl -L -o ~/elasticsearch-8.13.4-amd64.deb \
  https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-8.13.4-amd64.deb

# SCP from Mac to VM via jumphost
scp -J global.jumphost-prod.fkcloud.in \
  -i ~/.ssh/google_compute_engine \
  ~/elasticsearch-8.13.4-amd64.deb \
  <your-ssh-username>@<your-vm-internal-ip>:~/
```

Important: Run SCP from your Mac terminal, not from inside the VM. The VM is the destination.

Step 8: Install Elasticsearch on VM

```
sudo dpkg -i ~/elasticsearch-8.13.4-amd64.deb
# When asked about elasticsearch.yml → type N (keep our config)
# When asked about jvm.options → type Y (take new 8.x version)
```

Step 9: Configure Elasticsearch

```
sudo tee /etc/elasticsearch/elasticsearch.yml > /dev/null << 'EOF'
cluster.name: <your-cluster-name>
node.name: <your-cluster-name>-node1
path.data: /var/lib/elasticsearch
path.logs: /var/log/elasticsearch
network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false
EOF
```

Step 10: Fix data directory

```
sudo rm -rf /var/lib/elasticsearch
sudo mkdir -p /var/lib/elasticsearch
sudo chown -R elasticsearch:elasticsearch /var/lib/elasticsearch
```

Required if upgrading over a previous ES install — old state files from ES 5.x cause ES 8.x to fail with "Upgrading to 8.13.4 is only supported from version 7.17.0".

Step 11: Start Elasticsearch

```
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch
sudo systemctl restart elasticsearch
sleep 15
```

```
# Verify
curl http://localhost:9200
curl http://localhost:9200/_cluster/health?pretty
```

Expected: JSON with `"version": {"number": "8.13.4"}` and cluster status `green` .

Step 12: Test indexing

```
curl -X PUT http://localhost:9200/test-index

curl -X POST http://localhost:9200/test-index/_doc \
-H 'Content-Type: application/json' \
-d '{"name": "test", "status": "ES is working"}'

curl http://localhost:9200/test-index/_search?pretty
```

Phase 4 — Kibana Installation

Step 13: Download Kibana on Mac and SCP to VM

```
# Run on Mac
curl -L -o ~/kibana-8.13.4-amd64.deb \
https://artifacts.elastic.co/downloads/kibana/kibana-8.13.4-amd64.deb

scp -J global.jumphost-prod.fkcloud.in \
-i ~/.ssh/google_compute_engine \
~/kibana-8.13.4-amd64.deb \
<your-ssh-username>@<your-vm-internal-ip>:~/
```

Step 14: Install and configure Kibana

```
sudo dpkg -i ~/kibana-8.13.4-amd64.deb

sudo tee /etc/kibana/kibana.yml > /dev/null << 'EOF'
server.port: 5601
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://localhost:9200"]
EOF

sudo systemctl enable kibana
sudo systemctl start kibana

sleep 30 && curl http://localhost:5601/api/status | python3 -c \
"import sys,json; d=json.load(sys.stdin); print(d['status']['overall']['level'])"
```

Expected output: `available`

Step 15: Access Kibana from Mac via SSH tunnel

```
# Run on Mac – keep this terminal open while using Kibana
ssh -J global.jumphost-prod.fkcloud.in \
-i ~/.ssh/google_compute_engine \
-L 5601:localhost:5601 \
<your-ssh-username>@<your-vm-internal-ip> -N
```

Then open <http://localhost:5601> in browser. Press Ctrl+C in the tunnel terminal to close when done.

VM Summary (template — fill in your own values)

Property	Value
VM Name	<your-vm-name>
Internal IP	<your-vm-internal-ip>
Zone	<your-zone>
Machine Type	n2-standard-4 (4 vCPU, 16GB RAM)
Disk	100GB pd-ssd
OS	Debian 12 (Bookworm) — <golden-image-from-step-3>
Java	OpenJDK 17
Elasticsearch	8.13.4 — Running on port 9200
Kibana	8.13.4 — Running on port 5601

Common Errors and Fixes

Error	Cause	Fix
Permission denied (publickey) on SSH	Wrong username format	Use your Flipkart email with underscores instead of dots (e.g. <code>firstname_lastname_flipkart_com</code>), not the dotted form
WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED	Jumphost key rotated	<code>ssh-keygen -f "~/.ssh/known_hosts" -R "global.jumphost-prod.fkcloud.in"</code>
UseConcMarkSweepGC JVM error	ES 5.x + Java 17 incompatible	Use ES 8.x — do not attempt ES 5.x on Debian 12
Upgrading to 8.13.4 is only supported from 7.17.0	Old 5.x state files in data dir	<code>sudo rm -rf /var/lib/elasticsearch</code> then restart
Permission denied on /etc/default/elasticsearch	Wrong file permissions	<code>sudo chmod 644 /etc/default/elasticsearch</code>
Failed to locate elasticsearch-systemd-pre-exec	Old 5.x service file cached	<code>sudo systemctl daemon-reload</code>
SCP fails with "No such file" or connects to wrong host	SCP run from inside VM instead of Mac	Always run SCP commands from your Mac terminal, not from inside the VM

Reconnect Commands (template)

```
# SSH into VM
ssh -J global.jumphost-prod.fkcloud.in -i ~/.ssh/google_compute_engine <your-ssh-username>@<your-vm-internal-ip>

# Open Kibana tunnel (run on Mac, keep open)
ssh -J global.jumphost-prod.fkcloud.in -i ~/.ssh/google_compute_engine -L 5601:localhost:5601 <your-ssh-username>@<your-vm-internal-ip> -N

# Check ES health
curl http://localhost:9200/_cluster/health?pretty

# Check Kibana status
curl http://localhost:5601/api/status | python3 -c "import sys,json; d=json.load(sys.stdin); print(d['status']['overall']['level'])"
```

```
# Restart services
sudo systemctl restart elasticsearch
sudo systemctl restart kibana
```